Sustainable Housing

United Kingdom - Shawm House

Brazil – Subtraction House

Italy – House of Wood, Straw & Cork

Group 7 – D. Martini (s8097697), P. Deep Singh (s8076712), D. Caputo (s4528393)

Theme Selection

Sustainable Housing has been chosen as the group theme to enable each team member to look closely at an example of international residential construction. Initially, the theme allowed our members to demonstrate how sustainable methodologies have been incorporated internationally in building industries, however it also enables our team to compare these to those currently adopted in the Australian building industry. As awareness and knowledge grow, it is of increasing importance that the building industry collectively learns from each other and works together to design sustainable homes for our health and the future health of the planet.

Sustainable Housing - UK

Shawm House

(D. Martini s8097697)

Architect: Mawson Kerr

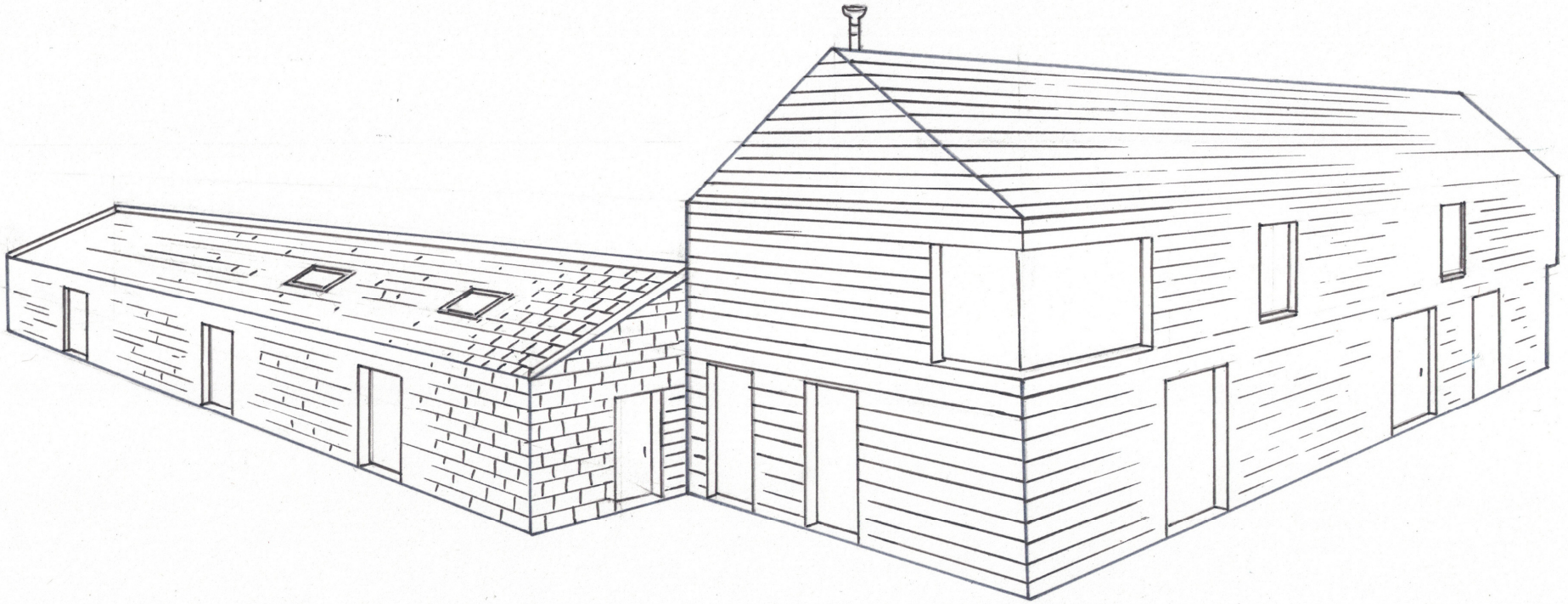
Size: 274 m2 Completed: 2017

Reflection

Deemed by the local council to be in the open countryside, Shawm House is located in the rural area of West Woodburn, Northumberland, United Kingdom. The project consists of a new build and the incorporation of an existing stone stable and stone wall. The stone stable was a listed building.

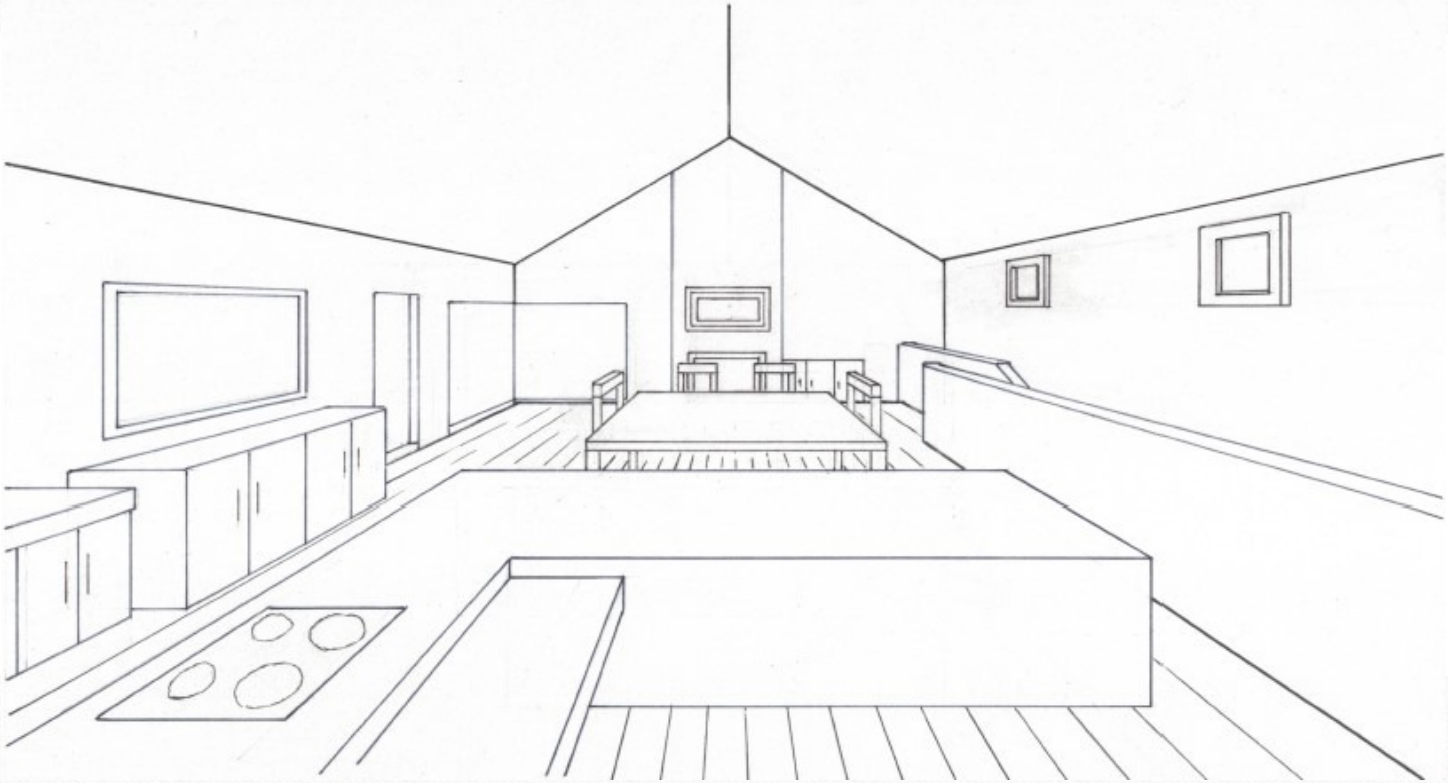
Passivhaus principles have been employed and the result is a well designed, highly insulated and airtight home. Featuring triple glazed windows and doors throughout, quality insulation, rainwater storage, a mechanical heat recovery ventilation system and a solar photovoltaic system. Materials have been sourced locally, reducing transportation costs and greenhouse gas emissions. The home also incorporates a small elevator which serves to future-proof the build for the retired owners.

Incorporating an existing building into the new design demonstrates an important consideration to both sustainability and sustainable housing - to preserve, not waste, existing structures and to reimagine for future use. The considered design has saved the structure from ruin while also reducing land fill.



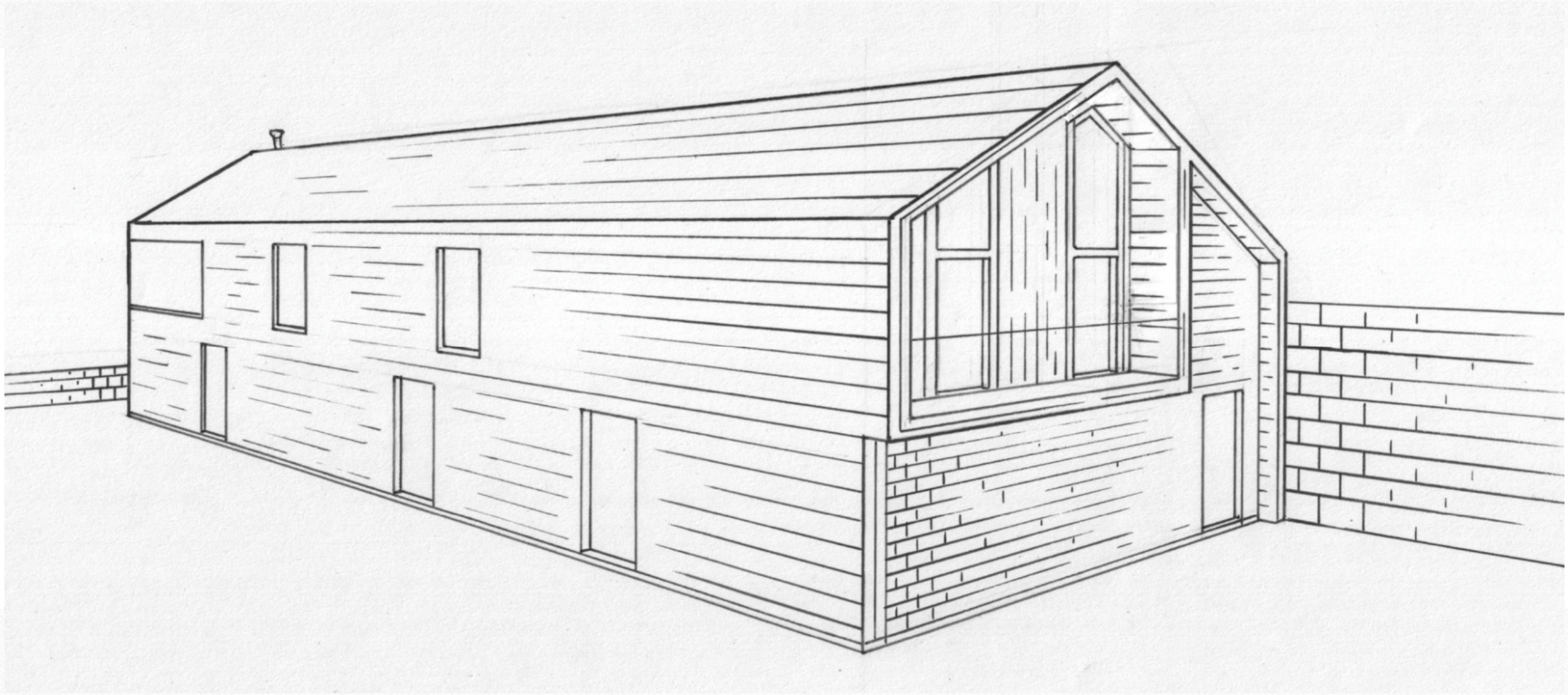
Hand Drawn External Perspective

North East View



Hand Drawn Internal Perspective

First Floor – East View



Hand Drawn External Perspective

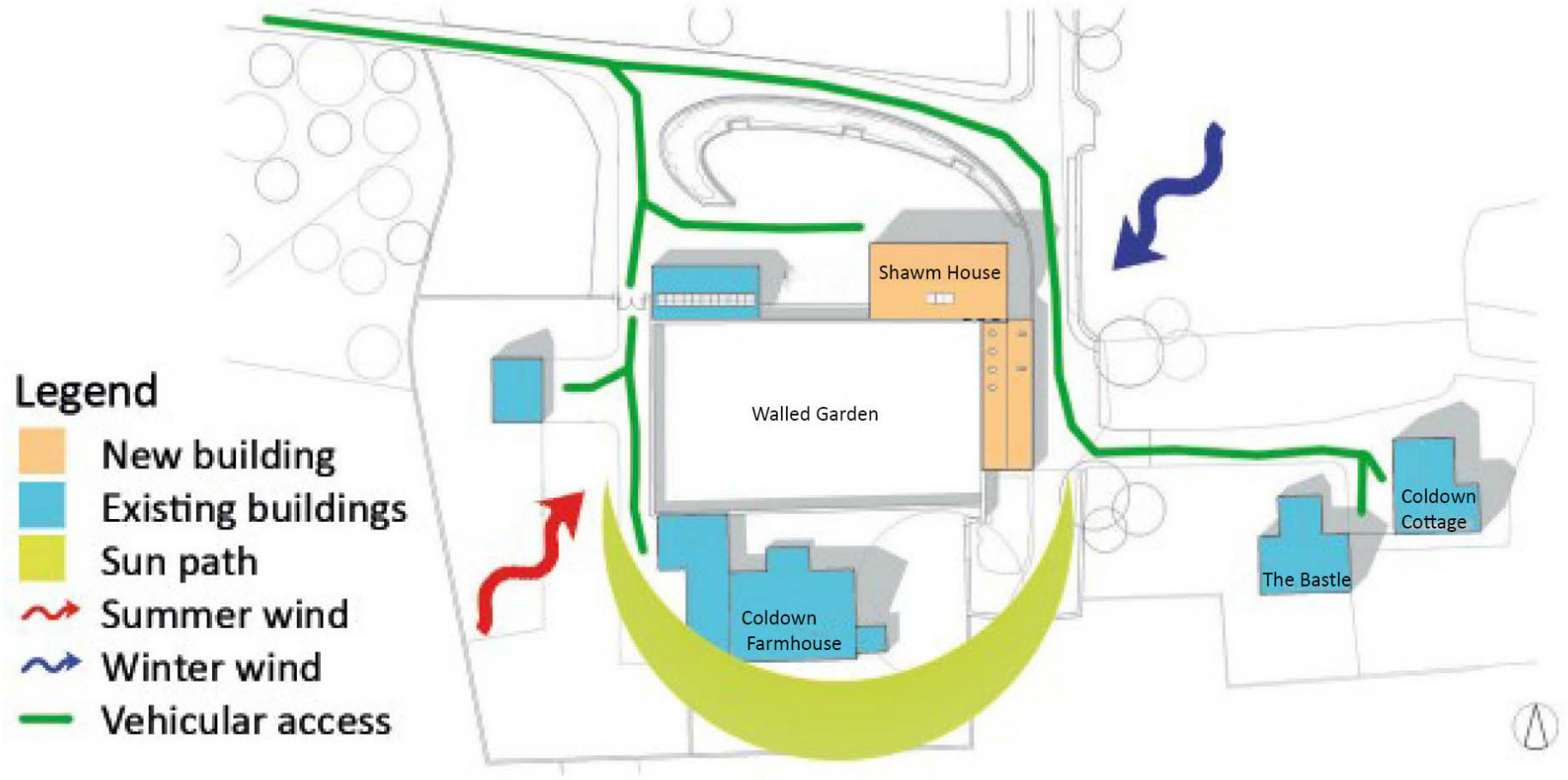
North West View

Site Analysis Diagram

In the following Site Analysis Diagram, the new build and existing buildings have been colour coded for quick identification. As indicated by the north direction arrow, in the new build two bedrooms on the ground floor, and the living areas and kitchen on the first floor, are on the north facing side. The glazing of these rooms makes use of natural daylight, minimising the need for artificial lighting. The kitchen has two west facing vertical windows and glazed doors however some protection is provided from the eave overhang and triple glazing. The stone stable has been reconfigured to house a bedroom, bathroom and lounge area. All feature glazing on the east side, while the lounge area also features a large south facing window.

The prevailing winter wind comes from the north-east, while the summer wind is from the south-west. The existing stone wall has been retained in the design and provides protection from prevailing winds.

Vehicular access routes are designed to provide access to each of the buildings on the site while also offering privacy to residents. The inclusion of stone gravel on the driveways enables stormwater to be absorbed, minimising run off.


Legend
New building
Existing buildings
Sun path
Summer wind
Winter wind
Vehicular access
Shawm House
Walled Garden
Coldown Farmhouse
The Bastle
Coldown Cottage

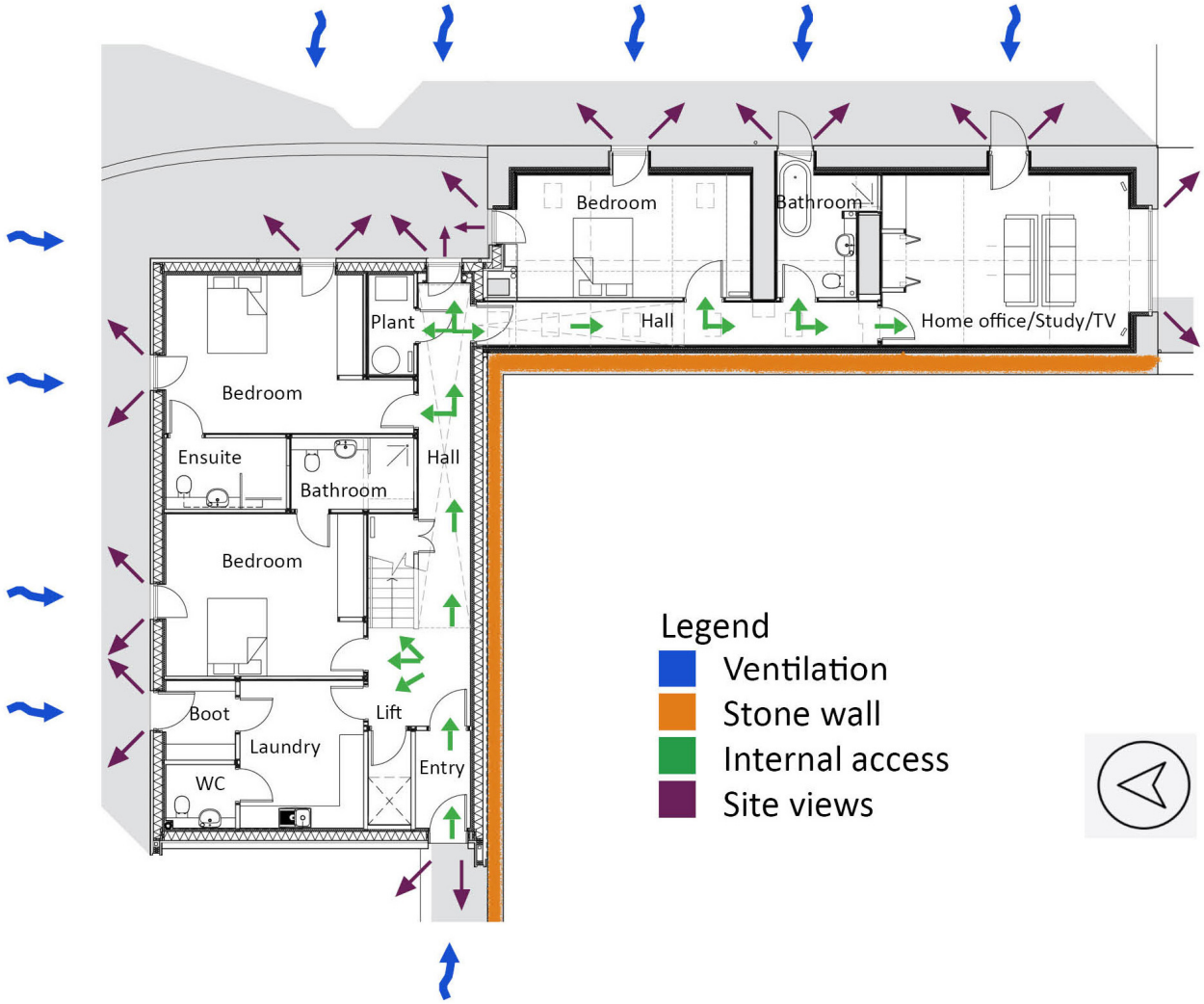
Floor Plan Analysis Diagram

In the following Floor Plan Analysis Diagram, cross ventilation is provided by the well positioned windows and doors throughout and the openable skylight windows in the roof of the stone stable and upper windows on the first floor of the new build. Inclusion of the skylights and upper windows enables any rising heat in the build to be purged at night.

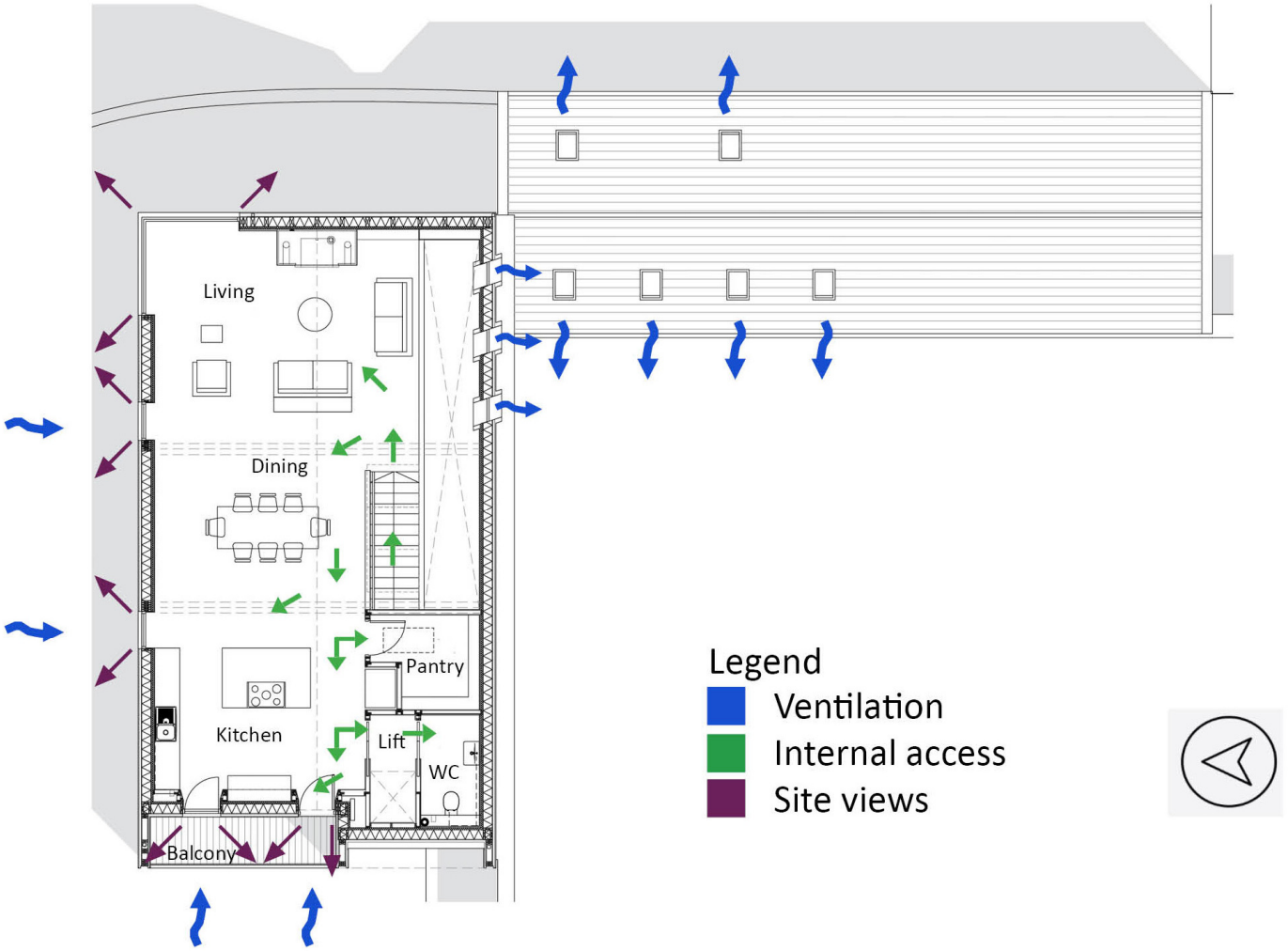
With the existing stone wall and the stone side of the stable, the build is protected from the prevailing winds and the occasional 30 deg Celsius high of summer, and 0 deg Celsius low of winter.

Access throughout the ground floor is via the hallway. Doors segregate the hallway into two parts, while the entry door to each room off the hallway provides privacy to each while also enabling noise control. Access to the first floor is via the staircase (or lift) and the open living area offers easy access to each room and the external balcony.

The majority of rooms in the new build and renovated stable face in the northern or eastern direction which affords each room a rural view and therefore overlooking the existing buildings on the site is avoided.



Ground floor


Living
Dining
Kitchen
Pantry
Lift
WC
Balcony
Legend
Ventilation
Internal access
Site views

First floor

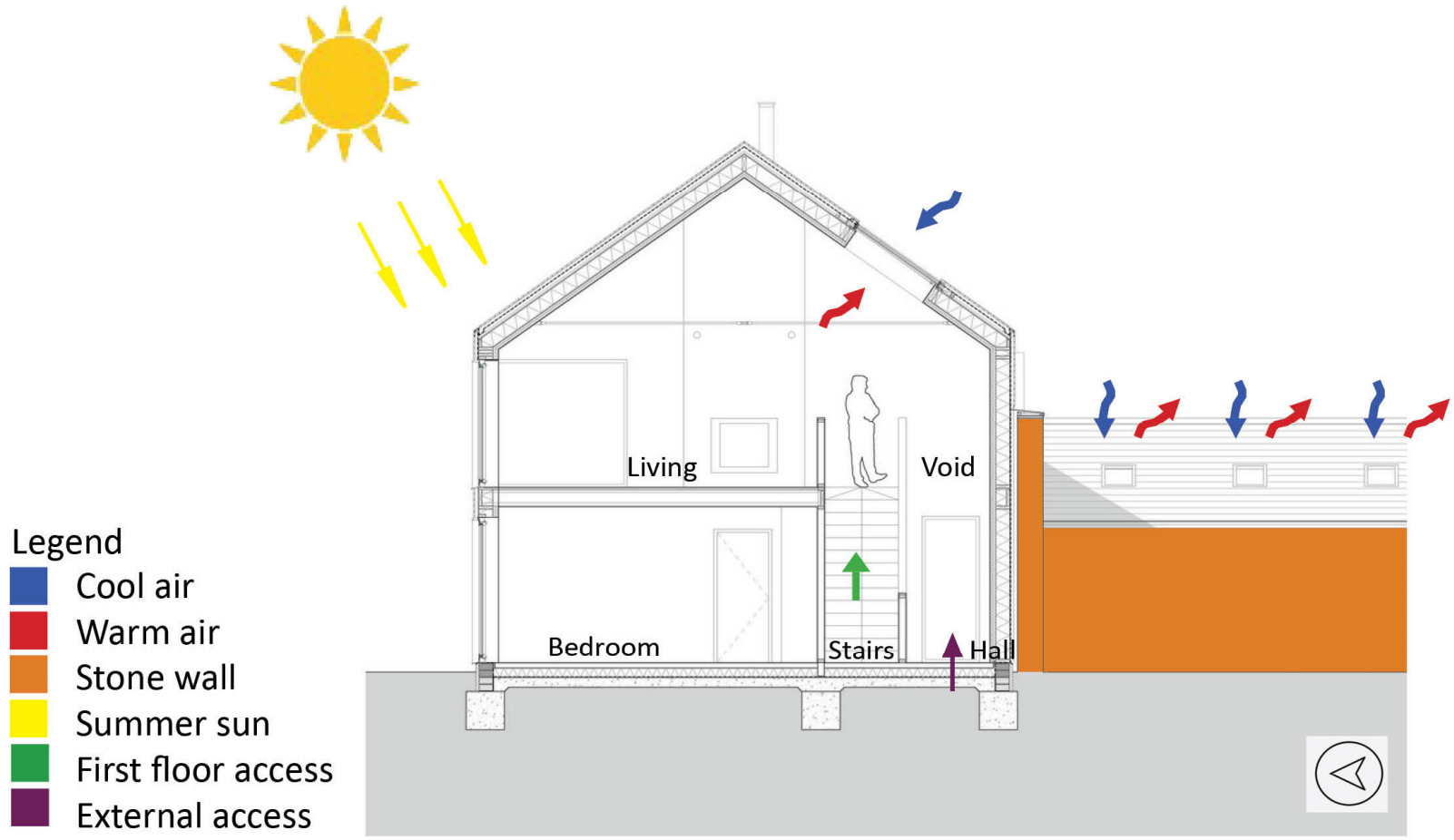
Vertical Section Analysis Diagram

In the following Vertical Section Analysis Diagram, the skylights and upper windows allow cross ventilation, providing an escape route for any rising heat. This enables the build to maintain a constant comfortable temperature for occupants.

The external existing stone wall and the stone wall side of the stable provide protection and insulation from the winds of the region.

The summer sun in the United Kingdom has a maximum height of approximately 61 deg during the summer solstice. While shading has not been used here, the design is considered and includes triple glazing and insulation throughout. Internally, the home has a constant temperature and natural lighting is a priority.

Access to the first floor is provided by both a staircase and a lift, which has been included to future proof the home for the retired owners. External access on the ground floor is predominantly via glazed doors which also serve to provide natural lighting into the rooms. Natural lighting and external access throughout the build has been well considered, with the inclusion of triple glazed doors.


Legend
Cool air
Warm air
Stone wall
Summer sun
First floor access
External access
Living
Void
Bedroom
Stairs
Hall

Model Making

Materials purchased

- Balsa wood, 50 x 300 x 3mm
- Balsa wood, 100 x 915 x 3mm
- Balsa wood, 100 x 915 x 1mm
- Match sticks
- Cork, 300 x 400 x 3mm
- Timber base
- Opaque paper

Tools

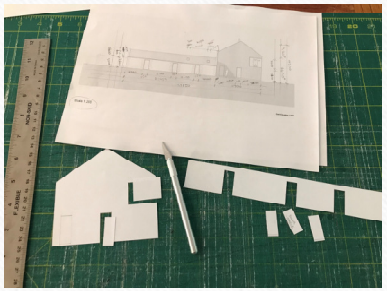
- Cutting mat, rulers, pencils, tweezers
- Scalpel, stanley knife
- uPVC glue, super glue, hot glue gun

- Step 1 – prepare scaled floor plans and elevations (completed 5/4)
- Step 2 – create wall, window and detail components from paper templates (completed 7/4)
- Step 3 – getting parts and roof ready for assembly (completed 11/4)
- Step 4 – cut and assemble internal components (completed 15/4)
- Step 5 – prepare model for final presentation (completed 16/4)

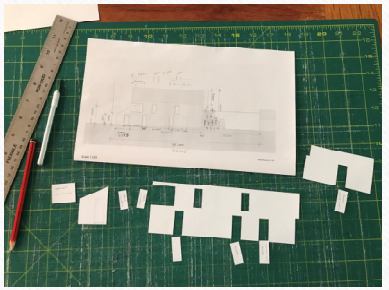
Model Making Progress

Materials and floor plans

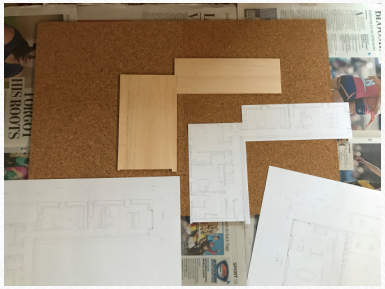
Materials and elevation drawings



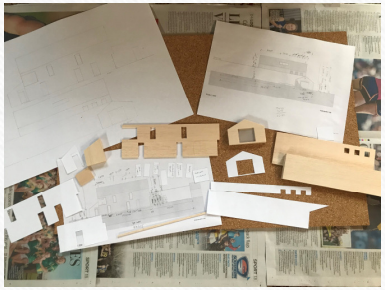
Cutting templates from drawings



Preparing templates



Preparing base and ground floor



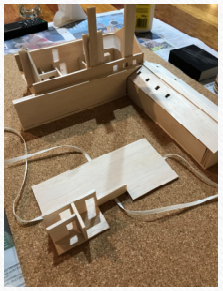
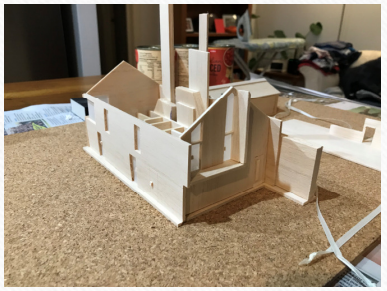
Cutting out elevation pieces

Cutting out external walls and details

Getting parts ready for gluing

Model Making Progress

Gluing and assembly of the model

Completed Model (Scale 1:100)

Group Poster Theme Style

- Size – A1
- Layout – portrait
- Font – Garamond
- Background – white with timber and nature images integrated
- Order -
    - Introduction, photo & reflection
    - Perspective drawings
    - Analysis diagrams & detailed information
    - Progress and final images of model

Conclusion

Designed according to Passivhaus principles, Shawm House is a great example of what is possible through early analysis, consultation with the client, and considered design. The build blends into the environment while taking advantage of a northern orientation and rural views, features carefully selected components and local materials, reinvigorates an existing heritage building and stone walls, and is designed to keep the occupants in their own home for as long as possible with low running costs.

As designers we can choose to design the home according to orientation and take advantage of wind direction. We can choose to recommend local materials rather than transporting materials from vast distances. We can recommend a high rated insulation and advise the best areas for it to be installed. We can also explain the 'whys' with our client prior to beginning the design process. The end result offers our clients a more comfortable and cheaper-to-run build that will serve them well now and into the future.

With knowledge, building sustainably is not complicated or difficult - it is a choice.

Sustainable Housing

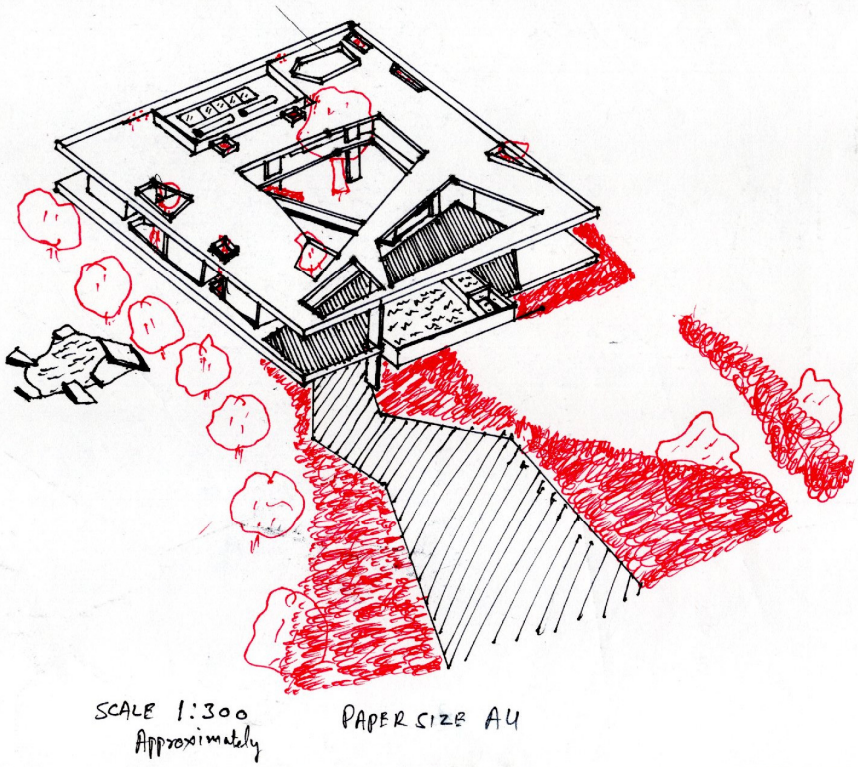
Subtraction House

Student ID: s8076712 (Pavan Deep Singh)

- Architects : Fernando Forte, Lourenço Gimenes, Rodrigo Marcondes Ferraz
- Area : 485 m2
- Construction Year : 2020
- Location : Ondominium Quinta da Baroneza, in Bragança Paulista, Brazil

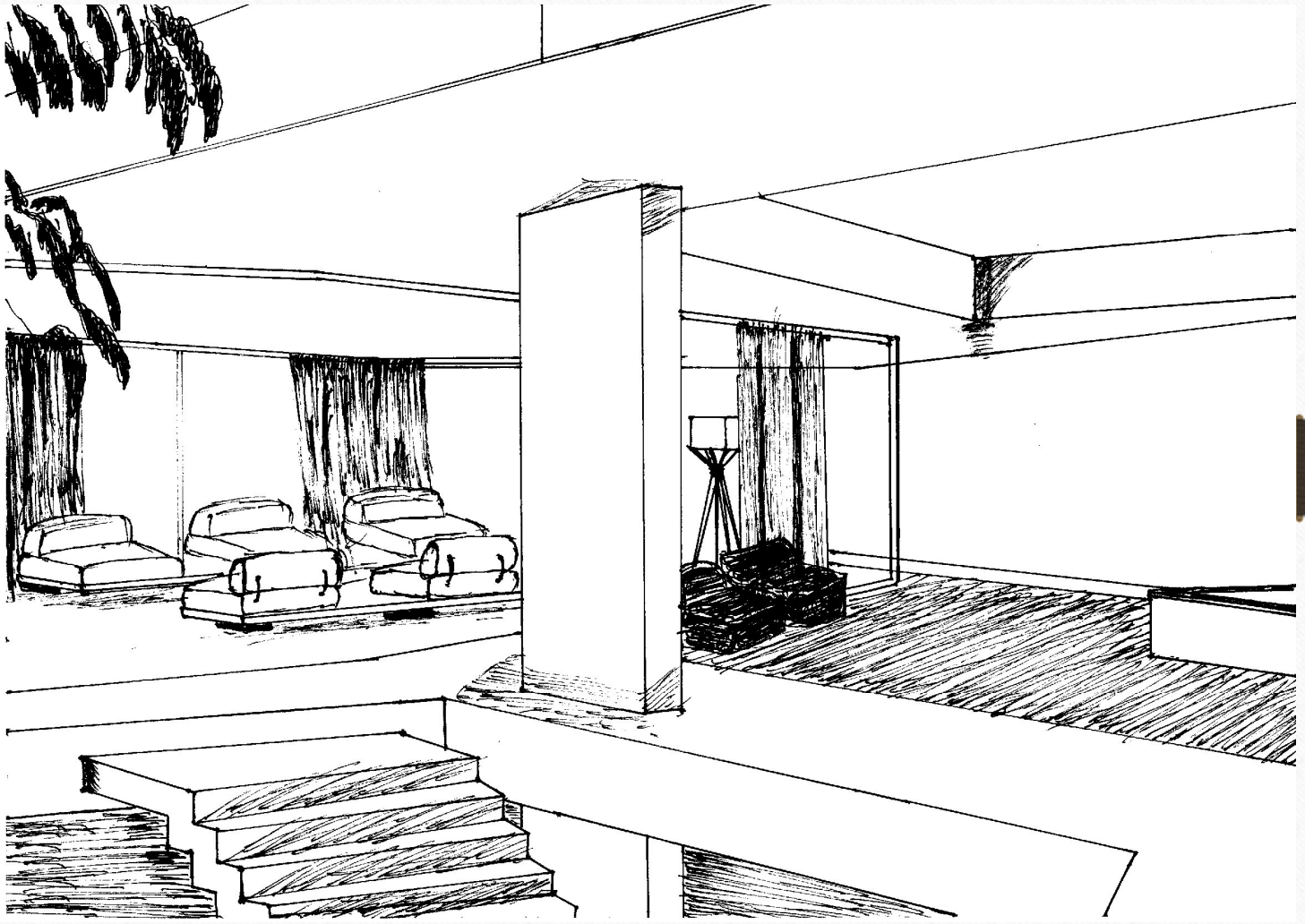
1.
Hand sketching
Stage 1

Subtraction House perspective drawing



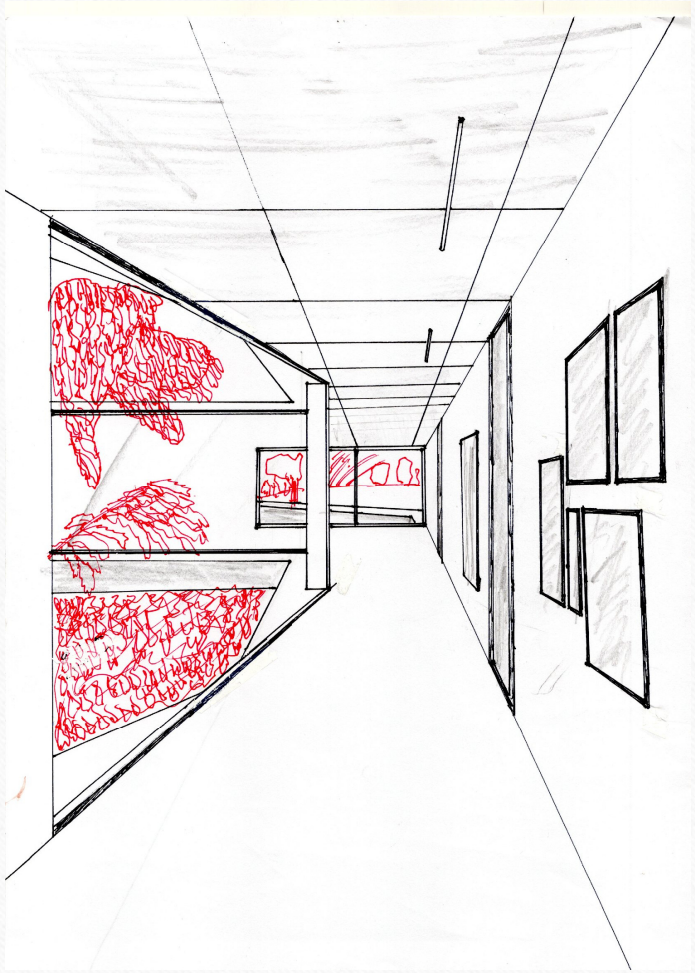
2.

Hand sketching
Stage 2



3.

Hand sketching
Stage 3

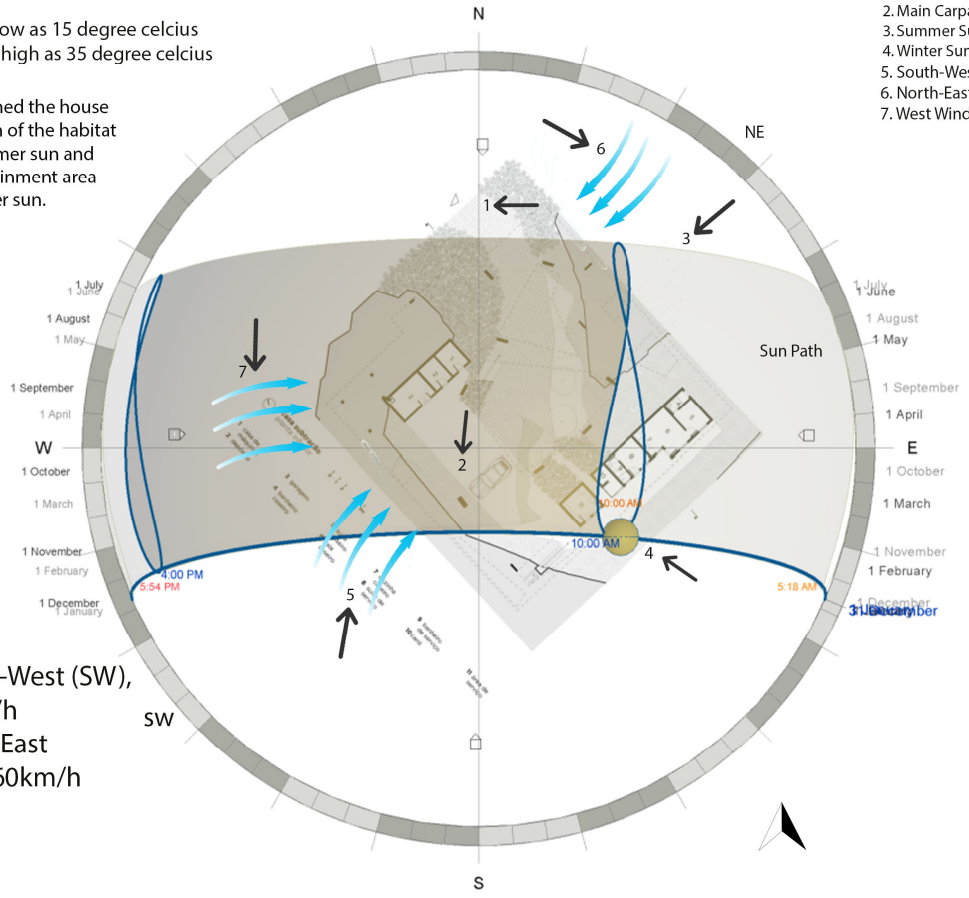


4. Site Plan Analysis

Subtraction House Designed by FGMF architects

Temperature in winter falls as low as 15 degree celcius
Temperature in Summer go as high as 35 degree celcius

FGMF architects perfectly designed the house with Sun path consideration non of the habitat place in the house face the summer sun and all the livable rooms and entertainment area take the full advantages of winter sun.

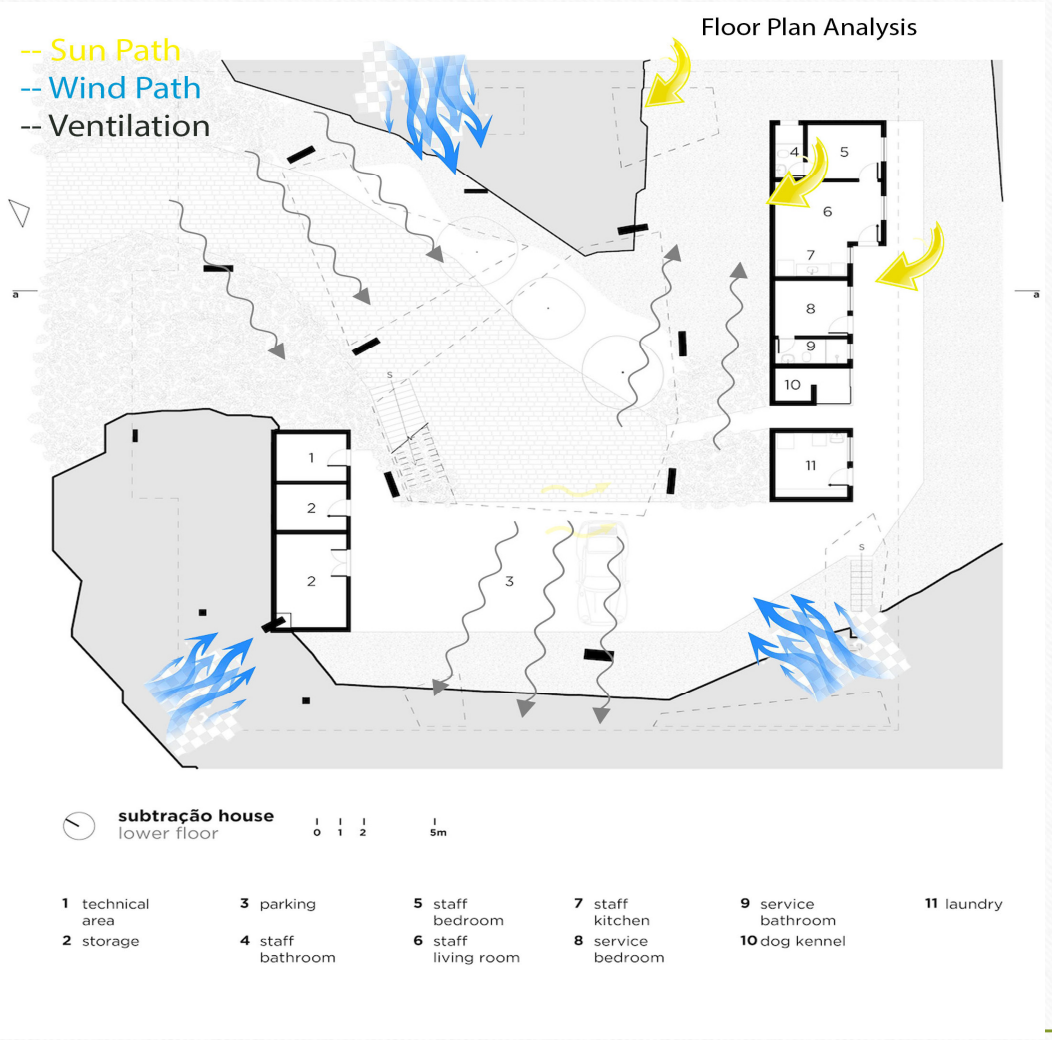


1. Site Access (Main Driveway)
2. Main Carpark
3. Summer Sun
4. Winter Sun
5. South-West (SW) Wind
6. North-East (NE) Wind
7. West Wind

Wind blows from South-West (SW), North-East (NE) <28km/h
Wind blows from West- East is the strongest with >60km/h

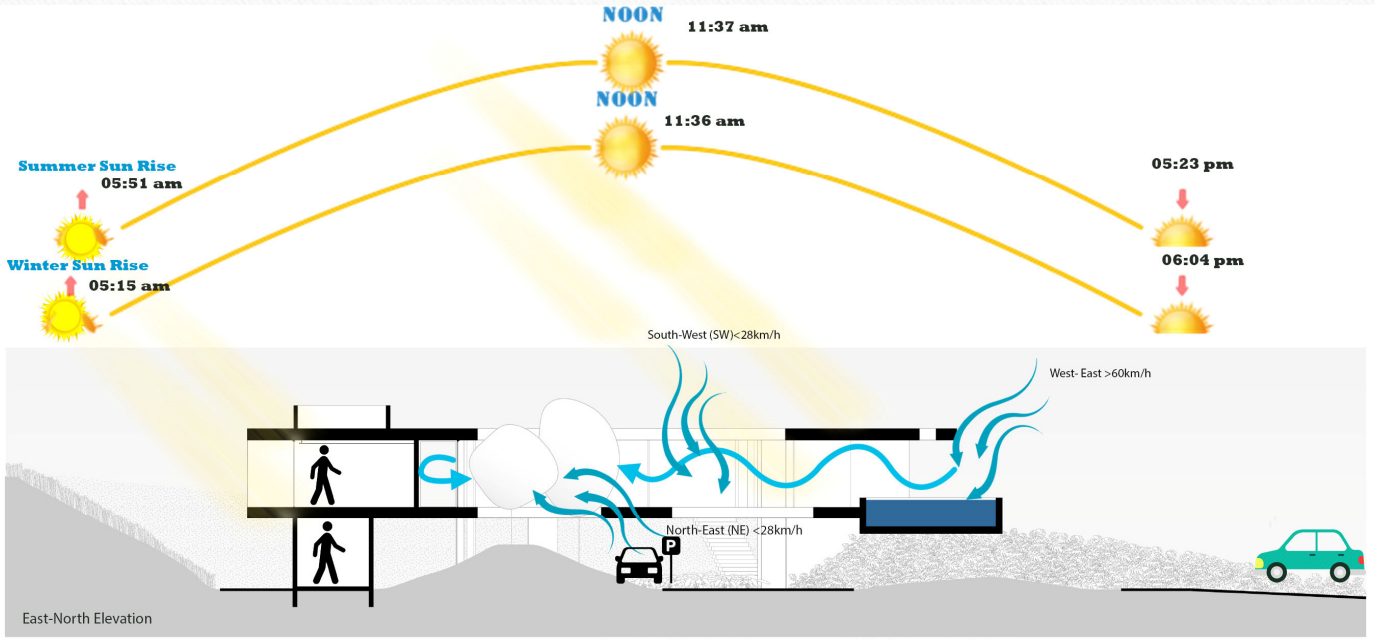
Scale 1:250

5. Floor Plan Analysis
Lower Floor



6. Floor Plan Analysis
Upper Floor

7. East North Elevation Analysis



0 1 3 5 10m

subtração house section a

Lowest Winter Temprature 15 degree celcius
Highest Summer Tempratute is 35 degree celcius

Wind blows from South-West (SW),
North-East (NE) <28km/h
Wind blows from West- East
is the strongest with >60km/h

Finished Floor to Main roof height is 7 Meter
Total Height of the Structure is 8 Meter

Sun Path
Wind Path
Ventilation
Site Access (Driveway)
Main Carpark
Bedroom

Notes on physical Model "Subtraction House "

8. Model making – plan

1. Start Printing the Plan
   31/03/24
2. Drawing on Balsa wood
   02/04/24
3. Cutting Shapes
   05/04/24
4. Starting to put it together
   09/04/24
5. Finishing
   15/04/24

Shopping list:

3mm Plywood	$17.00
20 x Plain unpainted Balsa wood DIY	$25.54
1 x Gorilla Glue Clear 51 ml	$13.60
1x Acetate Sheet (54cm x 50cm Roll)	$10..00
2 x Skinny Sticks 148x5mm, 80pk	$5.00
1 x TOLEDO 100MM 180 ° PROTRACTOR	$54.95
1 x Pendulum Jigsaw and Blades'	$64.95
1 x Slime	$16.00

Tools and utensils from previous project:

Ruler, pencils, eraser, sharpener, Circular saw for timber cutting

Execution Strategy :

First roof and upper-level plan is drawn on plywood with protractor and scale

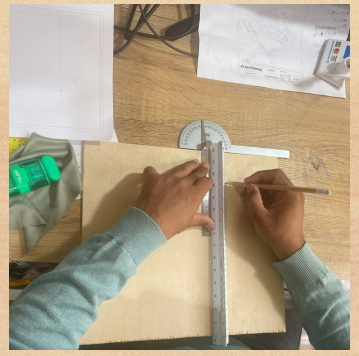
Then carefully every piece has been cut and measured for accuracy.

Gluing together all the structural parts

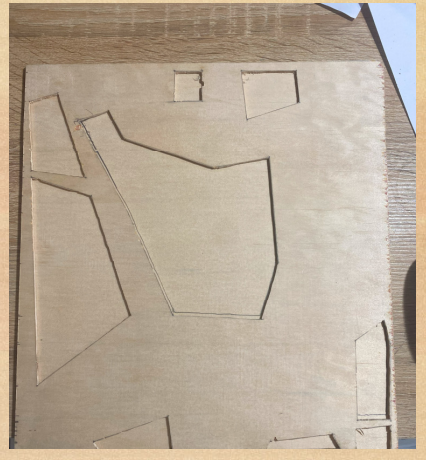
Acetate sheet has been cut and attached for the windows and open wall. of the model.

Final rendering is done, and model is finished.

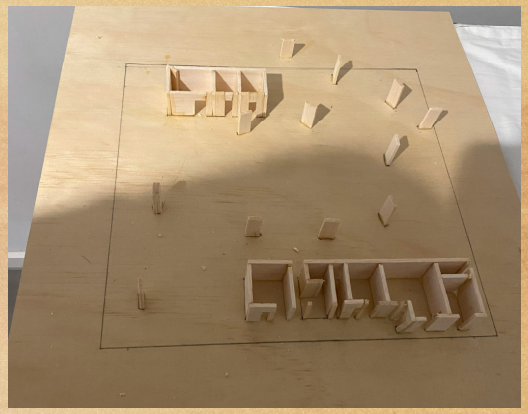
9. Physical Model Progress



Drawing the roof plan



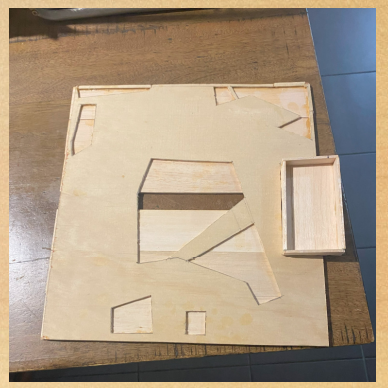
Cutting out roof plan



Walls elevated on ground floor

Cutting out upper floor plan

Cutting out swimming pool

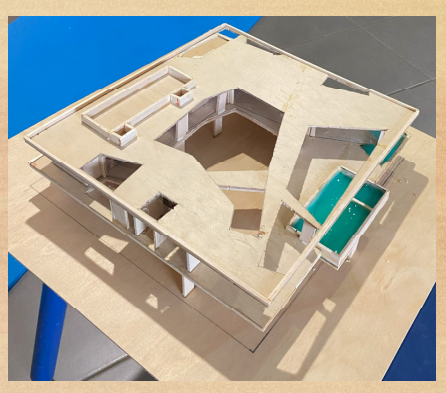


Putting together upper floor

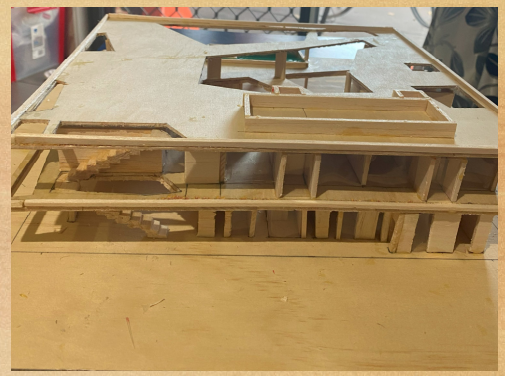
10. Final Model Presentation

Scale 1:100

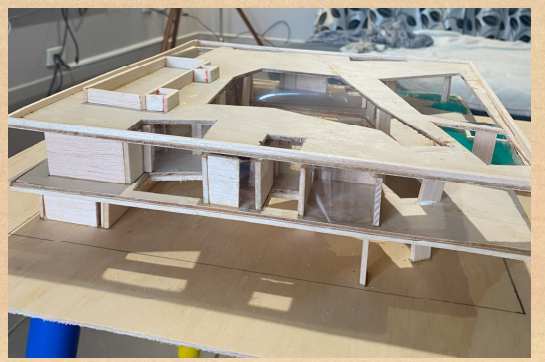
Drawing the roof plan



Cutting out roof plan



Drawing the upper floor plan



Cutting out upper floor plan

Putting together upper floor

Reflection

Our Group - 7, which consists of David, Debbie and myself decided to a research on sustainable housing around the world. It is more challenging to design and construct a sustainable house in developing countries. The main reason for that is the lack of training institutions, which could educate people about responsibly sourcing and use of sustainable materials and these institute could build a work force for this specific purpose in these countries.

I decided to pick this house "Subtraction House" in Brazil. Nearly half of Brazil covered in forests, using wood in construction is yet to be fully realized. This situation change the whole dynamics of sustainable housing. FGMF architects used perfectly brutalism philosophy, which prioritize functionality, honesty and social purpose. Even though this philosophy is fading away in the western world due to its bleak, cold, crude, and unattractive and unwelcoming look, the architect has done a remarkable job and change the whole perspective about the philosophy by using the original topography of the site to the maximum and incorporating all the existing tree in their design by making spaces in the house for them to flourish and enjoy the sun at the same time. This gave a warm and welcoming look to the house. In addition, the architects gave full consideration to sustainability and the house is built with recycled materials by incorporating recycled steel, glass, and concrete diverts waste from landfills.This remind us until full potential of wood and other sustainable materials can be realised, architectural brutalism will thrive in these developing counties.

There are some similarities among the group. However, they all have different designs, structure and construction and architectural philosophy.

Sustainable Housing

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The House of Wood, Straw and Cork

Designed by Luca Compri

- Our group has decided to choose the theme of sustainable homes as the idea of sustainability seems to be the future goal of construction. It includes locally sourced materials that will save on carbon emissions from exported products. My group has picked houses across the world with different ways of building with sustainable products, we will discover the similarities and comparisons of how each country has solutions to save on carbon emissions.
- I have picked my case study on a house located in a small town of Italy called Magnago. This home is made of very simple locally sourced materials with an extremely basic design. Luca Campri is the architect who created this sustainable home built in 2020. As the title depicts, the house is made from mostly wood, straw, cork and stone.
- We will dive into the structure of this home and how it saves on carbon emissions with its fully glazed windows and locally sourced materials. This home also is self powered as it is equipped with solar panels. If this home is ever decommissioned it will be easily recyclable from the natural materials involved.

- Located in Magnago, Italy
- Built in 2020
- Made of wood, straw and cork

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Plan for making my model

Model will be 1:100 scale

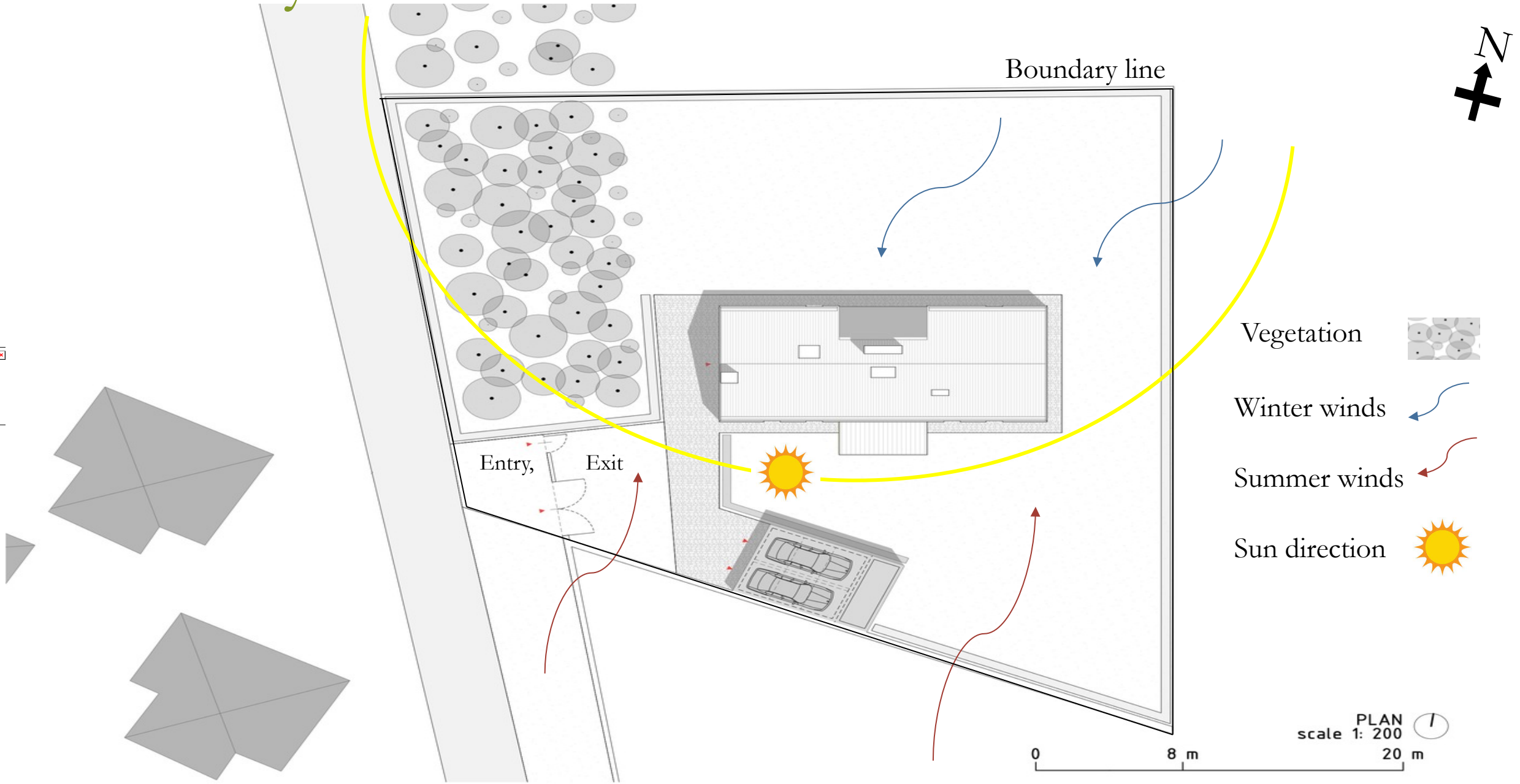
Materials needed-

-Will be sourced from spotlight
-Mainly using Balsa wood, acrylic sheet, foam/cardboard, glue, cutting tools, rulers, pliers -

Plan-
-To start my model I will be collecting all materials needed to start starting with the base for my model to sit on. The base will either be a foam or cardboard for a simple flat ground.
-I will be deciding the correct to scale to make this model (completed on 29th of march)
-I will then start by cutting and gluing balsa wood together to start creating this project, using acrylic for the windows and start looking at other materials to add for colour and to symbolise different materials used on the house.
Materials and model making will commence on the 1st of April

Cost analysis-
Balsa wood-$8 a sheet, estimate (3 sheets) $24
Glue – already own
Acrylic sheet $7 (1 sheet)
Cardboard base (1 sheet) $12

Site Analysis



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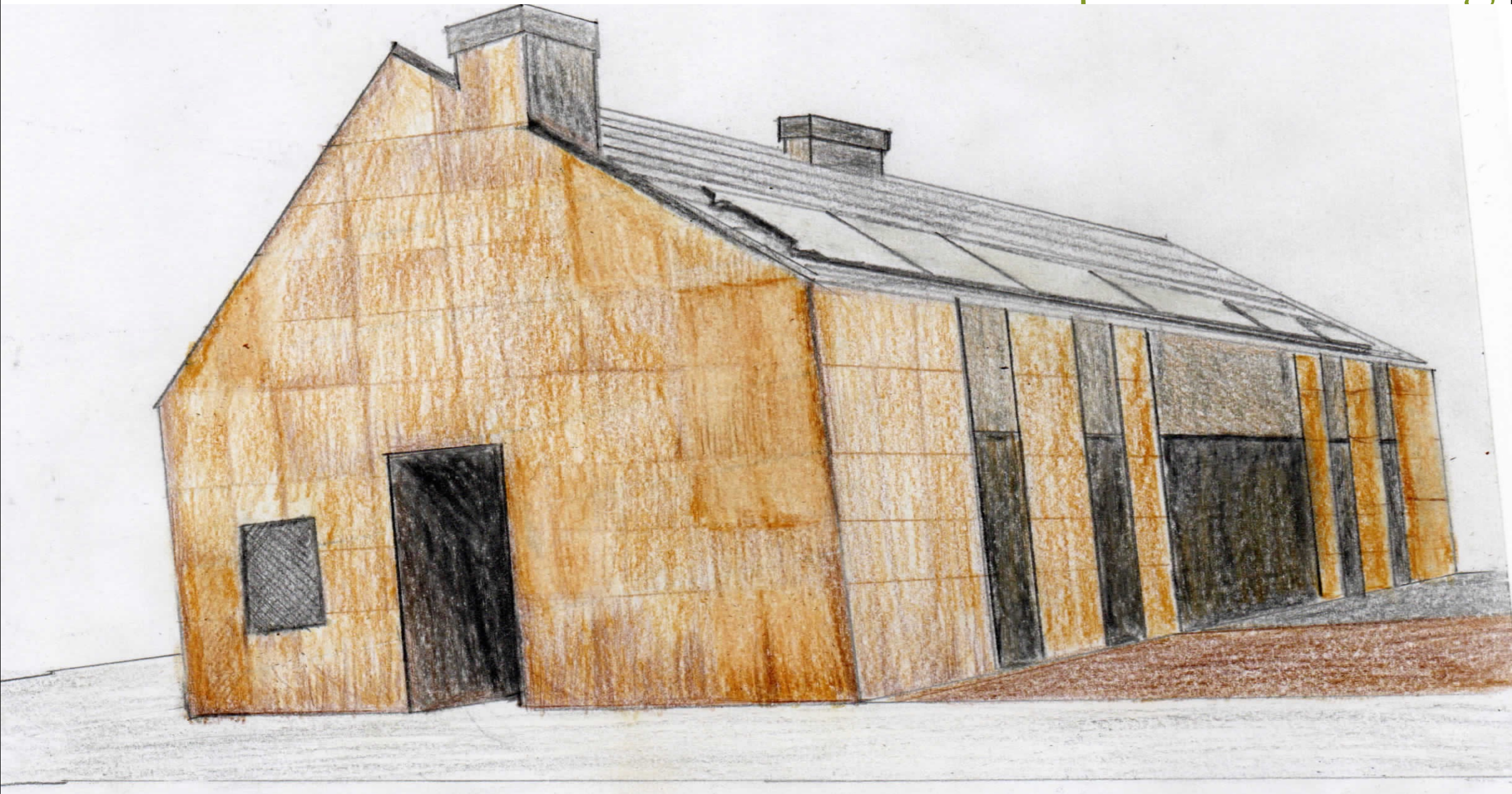
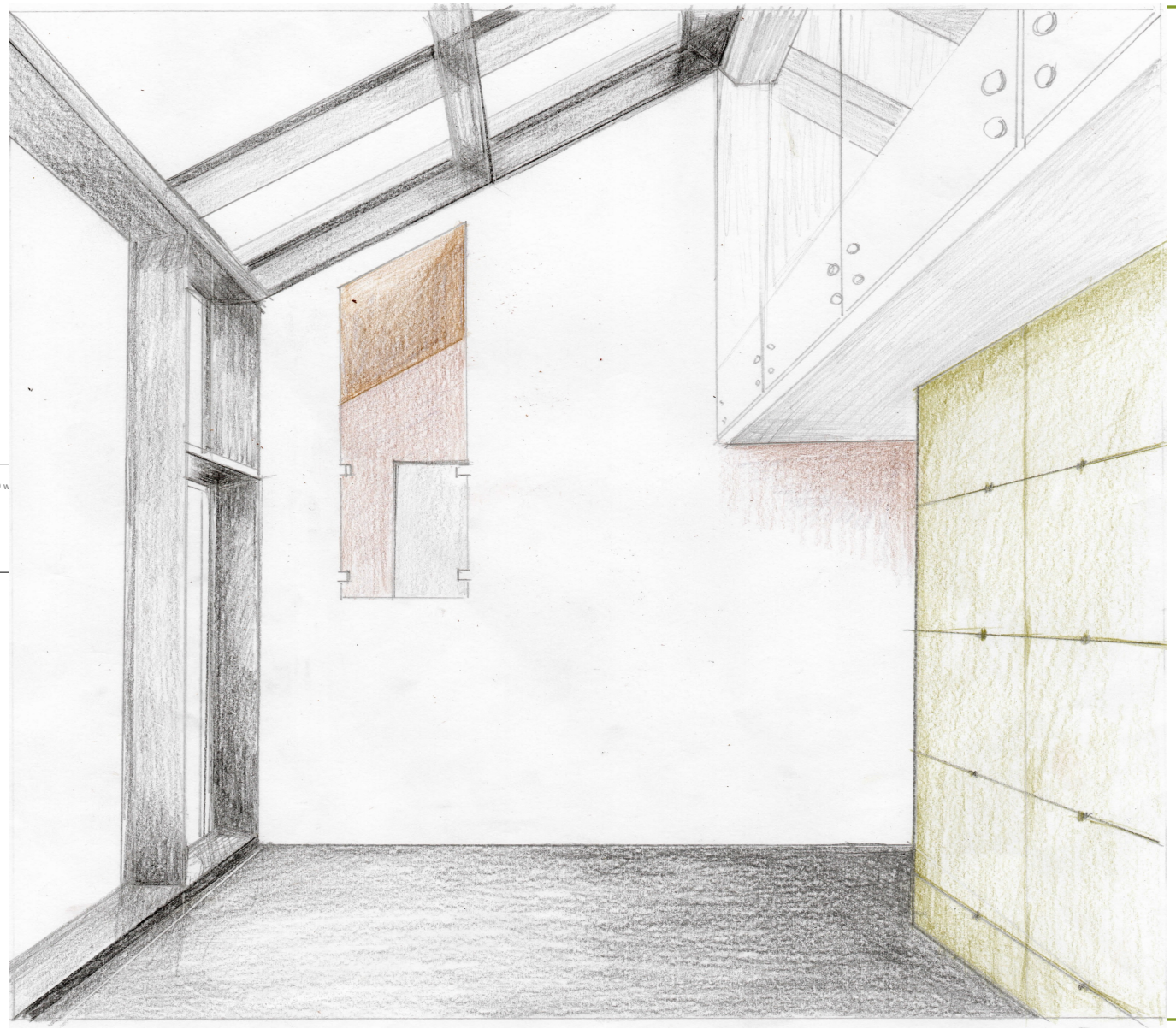
Perspective drawing



Perspective drawing 2



Perspective drawing 3

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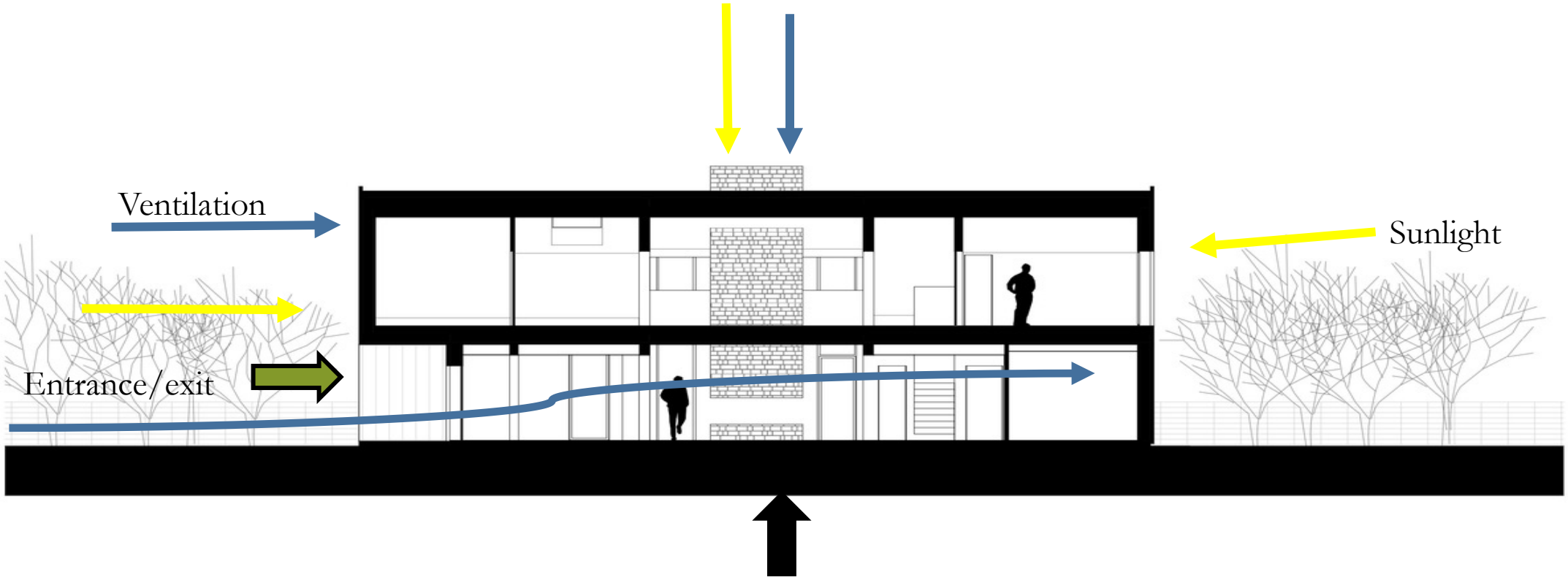
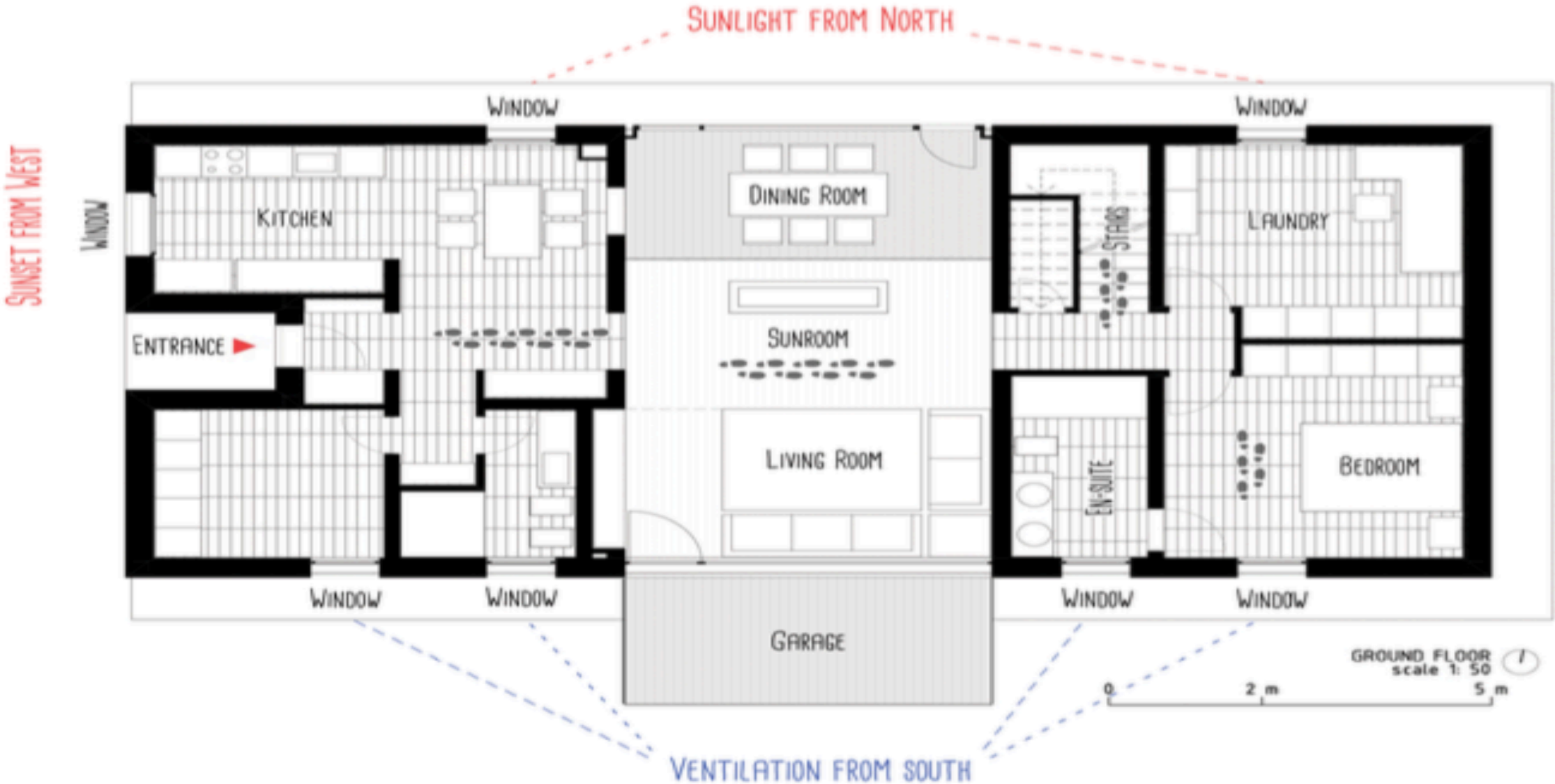
Vertical Analysis

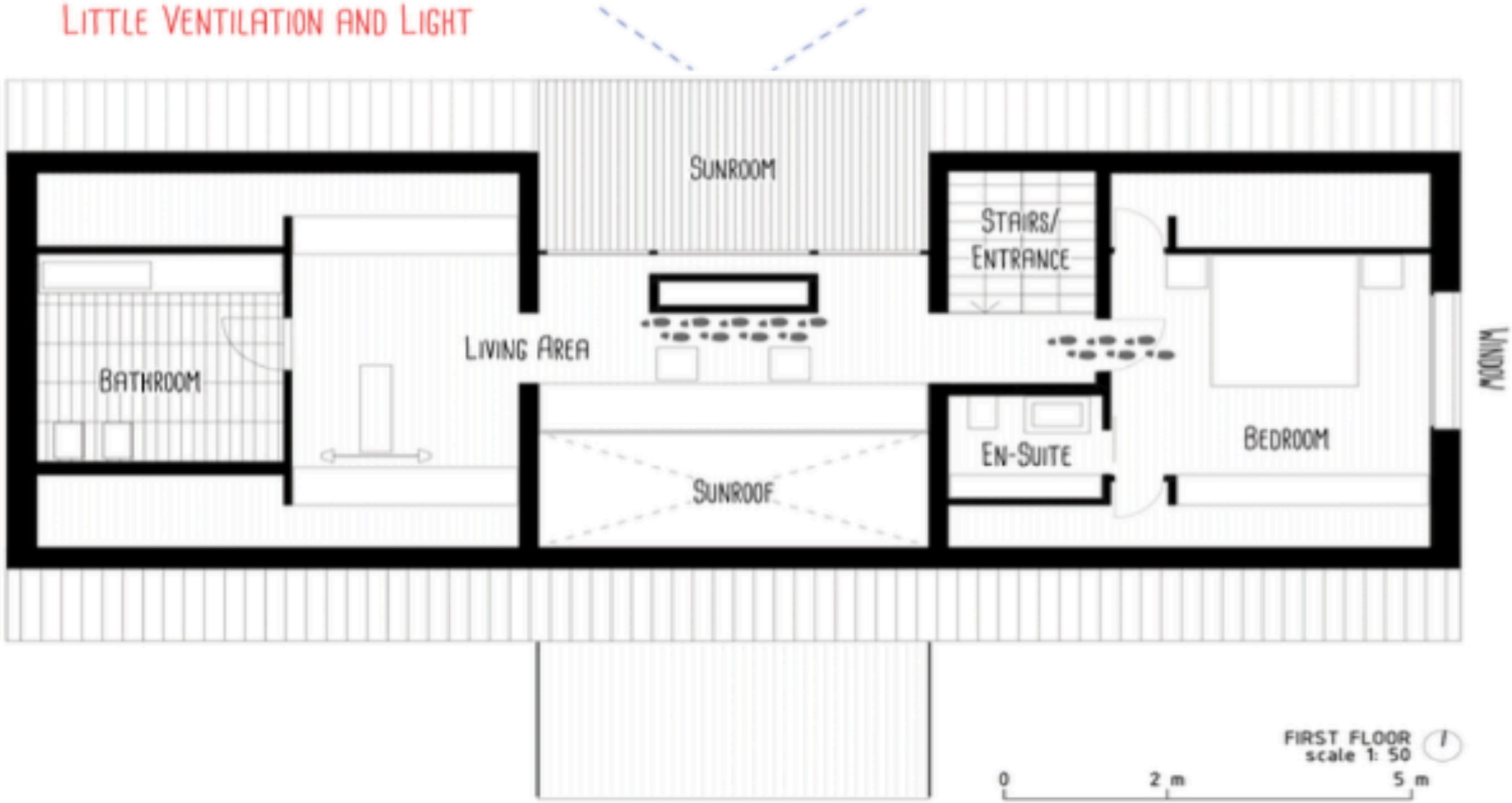


FLOOR PLAN ANALYSIS

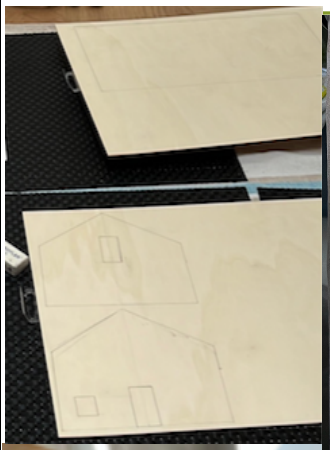


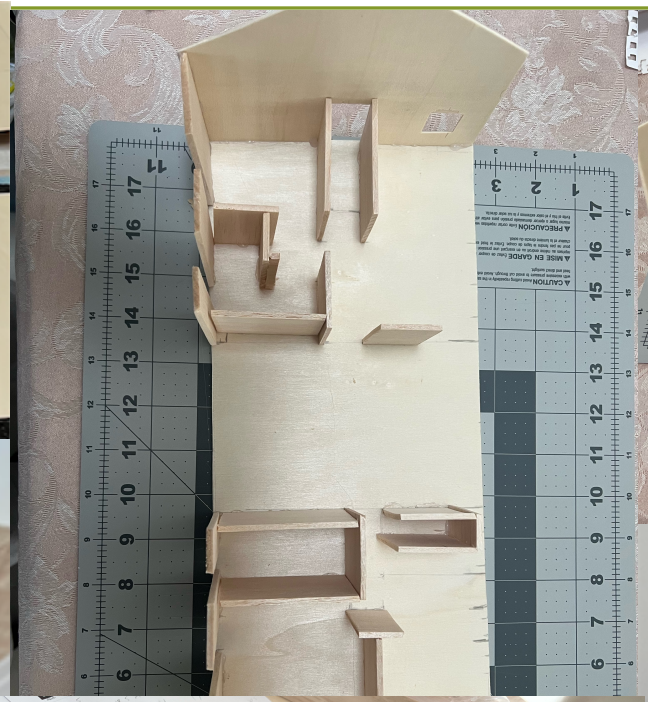
- FOOT TRAFFIC

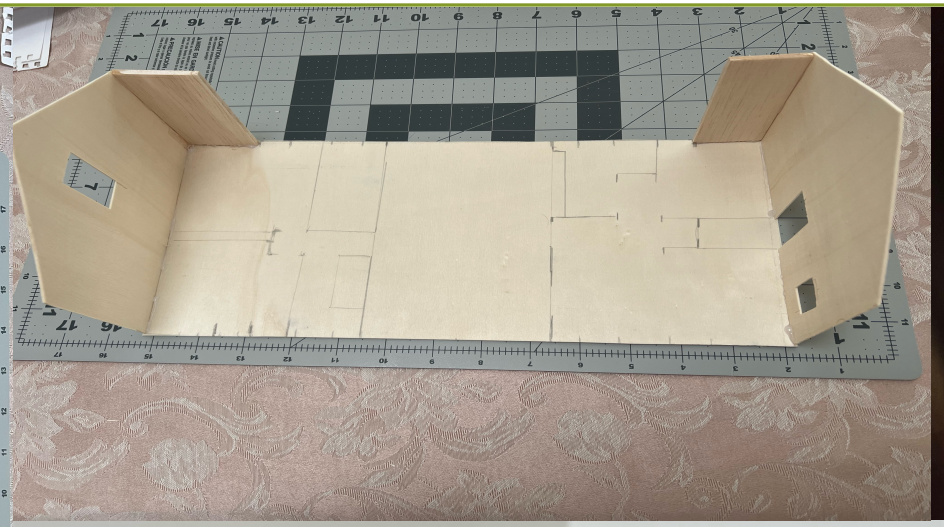
Little Ventilation and Light

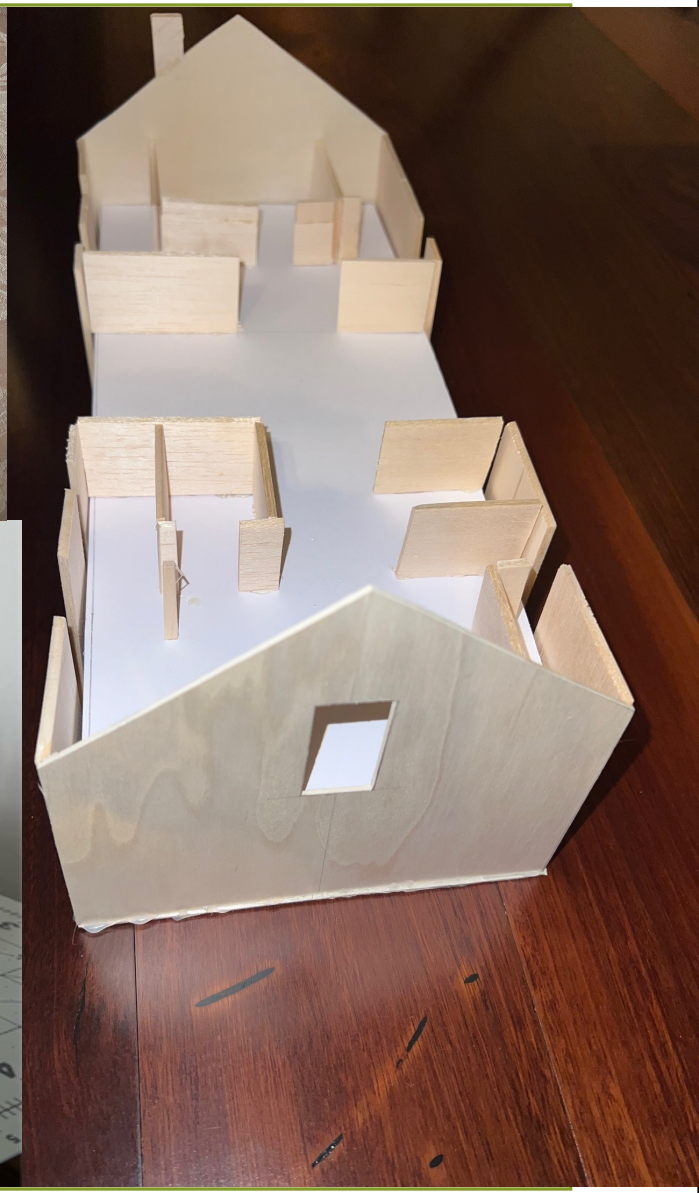


Model progression up to 50% completion









Reflection of group theme and case study

What we can gather from this project is it's very well designed to promote sustainability. This can be demonstrated from the first analysis, as we can see the sun room is facing the suns direction. This promotes light and passive heat throughout sunrise to sunset. This house has windows in certain points to let in warm air and cold air depending on season. The windows are placed in points of the home that are most used as the bedrooms or living areas. This property receives winter winds from the north and summer winds from the south. If we look into the floor plan analysis we see where most of the foot traffic is, these areas are surrounded by windows and ventilation to keep the residents comfortable with also reducing costs on AC and HVAC. This home has been built perfectly with sustainability in mind however as we can see in the floor plan analysis on the first floor near the bathroom there is no windows which suppresses light and ventilation. Otherwise this house is built very simply with mostly wood and cork with a plain design, this is an example of sustainability that is not costly in which can be replicated to create simple and efficient homes. I believe this group theme we have chosen is intriguing and has taught me a great deal of knowledge of the effects of natural sunlight, wind direction and ventilation.

Group Poster Theme Style

Size – A1Layout – portrait
Font - Garamond
Background – white with timber and nature images integrated

Order
Introduction & photo
Reflection
Perspective drawings
Analysis diagrams & detailed information
Progress and final images of model

Completed Model

Scale 1:50